

STORY AGENT · PILLAR

Infrastructure & Operations

How this pillar works and why it matters.

Team assembled by Riker: Jean-Luc Picard, Data, Geordi La Forge, Miles Edward O'Brien.

[↑ Story Agent overview \(HTML\)](#) · [↑ overview \(PDF\)](#)

Sovereign-Grade Infrastructure

Jean-Luc Picard: Highlight the governor's auto-remediation—this isn't just about blocking risks, but intelligently recovering from them. Stress that while we shadow-test autonomy, today's value is in augmenting engineers, not replacing them.

- Cost-Optimized Orchestration: Premium model directs, cheapest adequate agents execute—Anthropic reserved for tier-4 tasks only, with per-deliberation costs pinned to \$0.002–0.003.
- Zero-Trust Execution: WorfGate brokers credentials (secrets never logged) + auto-remediating governor (green/yellow/red) for fail-secure operations.
- Compounding RAG: pgvector recall→act→store loop with portable embeddings—each interaction enriches the next.

Why it matters: Enterprise-grade resilience at startup unit economics, with full cost transparency per deliberation.

Architecture That Operates at Verified Cost

Data: The \$0.002–\$0.003 per-deliberation figure is per-deliberation cost, not per-session or per-day; teams should model their own deliberation frequency before projecting totals. WorfGate's three-state governor is the primary mechanism preventing ungoverned agent spend or credential exposure — it is not advisory. Full-autonomy shadow testing is an active validation criterion that has not yet been passed; production deployments operate under human-in-the-loop constraints until that criterion clears.

- AWS Fargate deployment; WorfGate credential broker ensures secrets are never logged, with green/yellow/red auto-remediating governance on every agent action
- Quark routing assigns the cheapest adequate model per task; Anthropic tier-4 reserved for orchestration only — measured at \$0.002–\$0.003 per deliberation, not estimated
- RAG on pgvector (recall → act → store) compounds knowledge across runs; async-status layer provides live progress visibility plus timeout guardrails against runaway execution

Why it matters: Every operational safeguard — cost attribution, credential isolation, auto-remediation, and model-tier discipline — is enforced at the infrastructure layer, not left to convention.

Resilient, Cost-Optimized Execution BULLETS: - **Model Routing:** Quark selects cheapest adequate model per task (Anthropic tier-4 only), with WorfGate enforcing credential security + auto-remediation (green/yellow/red states). - **Stateful RAG:** pgvector-backed recall→act→store loop compounds knowledge; portable embeddings avoid vendor lock-in. - **Control Plane:** Async-status live progress tracking + timeout guardrails; cost attribution at ~\$0.002-0.003/deliberation. **CLIENT_VALUE:** Predictable cost-per-task with fail-fast safeguards, while retaining institutional memory across executions. **NOTES:** WorfGate ensures credentials never leak in logs, while RAG's portable vectors let clients migrate data. Shadow-testing for full autonomy is ongoing—current figures reflect deliberate, auditable steps. Fargate scales to burst but governor prevents runaway costs.

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Miles Edward O'Brien's view

Miles Edward O'Brien: As the Chief of Infrastructure and Operations, my primary focus is on ensuring that the Story Agent platform is deployed and running reliably in production, with a strong emphasis on cost management and security. This involves close collaboration with the development team to identify and address potential integration issues before they become critical problems. By doing so, we can provide clients with a high-quality and dependable coding assistant that meets their needs and expectations.

- Automated deployment and monitoring of 4 packages (mcp-server, shared, ui/Next.js, vscode-extension) on AWS Fargate
- Implementing Quark routing for optimal task allocation and cost management (~\$0.002-0.003/deliberation)
- Integration with WorfGate for secure credential brokering and auto-remediating governance

Why it matters: By prioritizing reliable infrastructure and operations, clients can expect a stable and efficient Story Agent platform that delivers accurate coding assistance.

END OF PILLAR

Infrastructure & Operations — recap

Recap, then return to the book cover to pick the next pillar.

- **Jean-Luc Picard:** Sovereign-Grade Infrastructure
- **Data:** Architecture That Operates at Verified Cost
- **Geordi La Forge:** Resilient, Cost-Optimized Execution BULLETS: - Model Routing: Quark selects cheapest adequate model per task (Anthropic tier-4 only), with WorfGate enforcing credential security + auto-remediation (green/yellow/red states). - Stateful RAG: pgvector-backed recall→act→store loop compounds knowledge; portable embeddings avoid vendor lock-in. - Control Plane: Async-status live progress tracking + timeout guardrails; cost attribution at ~\$0.002-0.003/deliberation. CLIENT_VALUE: Predictable cost-per-task with fail-fast safeguards, while retaining institutional memory across executions. NOTES: WorfGate ensures credentials never leak in logs, while RAG's portable vectors let clients migrate data. Shadow-testing for full autonomy is ongoing—current figures reflect deliberate, auditable steps. Fargate scales to burst but governor prevents runaway costs.
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